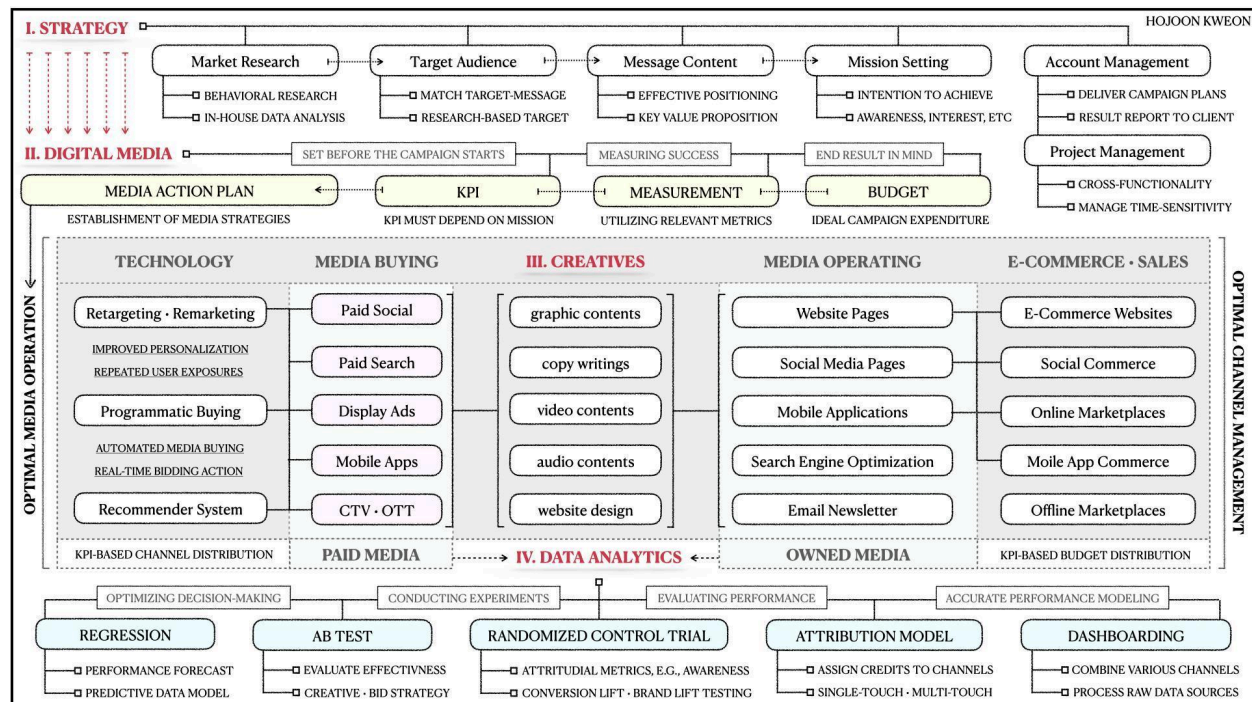


OBJECTIVE DIAGRAM



DATASET COLLECTION

CAMPAIGN KPI [TIME SERIES]: AWARENESS OVER CAMPAIGN DURATION

- Awareness of Campaign: Overall Awareness Over Campaign Duration
Link: Over a time period of a campaign (time series), what was the overall awareness of the overall campaign using the four channels? This awareness measurement should be derived from experiments of surveys, meaning that it is based on a scale (e.g., [0, 1] or [1, 9]). Based on this, we are finding beta for each channel contributing to awareness.

MEASUREMENT KPI: AWARENESS PER CHANNEL OVER CAMPAIGN DURATION

- Channel #1 - Instagram: Impression Metrics Over Campaign Duration
Link: <https://www.kaggle.com/code/thabresh/eda-marketing-campaign-performance/input>
- Channel #2 - YouTube: Impression Metrics Over Campaign Duration
Link: <https://www.kaggle.com/code/thabresh/eda-marketing-campaign-performance/input>
- Channel #3 - Search: Impression Metrics Over Campaign Duration
Link: <https://www.kaggle.com/code/thabresh/eda-marketing-campaign-performance/input>
- Channel #4 - Display: Impression Metrics Over Campaign Duration
Link: <https://www.kaggle.com/code/thabresh/eda-marketing-campaign-performance/input>

MATHEMATICAL APPROACH

- $\Delta \text{Mission KPI}(t) = (\beta_1)\Delta \text{Instagram}_t + (\beta_2)\Delta \text{YouTube}_t + (\beta_3)\Delta \text{Search}_t + \epsilon_t$
- **Interpretation:** A channel with higher beta (coefficient) will represent the most relevant contribution to awareness lift per 1 impression. A channel with lowest beta will represent the least relevant contribution to awareness lift per 1 impression.
- **Next Step:** In a campaign, we measure the awareness-contribution ratio of channels per 1 impression (e.g., 0.3 Instagram : 0.5 YouTube: 0.9 Search : 1.0 Display).
- **Next Step:** Using this contribution ratio, we calculate the weight ratio of causing the same amount of contribution to awareness (e.g., 100 impressions on Instagram = 10 views on YouTube = 150 impressions on Display Ads = 60 CTR on Google Search).
- **Next Step:** Using the weight ratio, we can calculate the qualitative ROAS where we can say \$1 investment on Instagram contributes by x amount, on YouTube by x amount, on Display Ads by x amount, and on Google Search by x amount to the campaign mission objective (e.g., 100 Instagram = 12 YouTube = 30 Search: 300 Display)
- **Application:** Based on qualitative ROAS per \$1, increase budget allocation on channels with higher qualitative ROAS, while decreasing budget allocation on channels with lower qualitative ROAS.
- **Further Research:** Obviously, it is not most effective to allocate all budget to a channel with highest qualitative ROAS, while removing all spendings from channels with lower qualitative ROAS. Fundamentally, marketing communication campaigns are most effective when multiple channels are integrated together. What would be the threshold where we can improve performance by increasing/decreasing budget allocation by applying the qualitative ROAS findings? This would be the follow-up question.

Project: Awareness over campaign

Methodology: Multiple Linear Regression (OLS) & Qualitative ROAS Calculation

1. Executive Summary

This analysis quantifies the contribution of four major digital channels based on 200,000 observations. The results indicate that Google Ads yields the highest awareness lift per unit of impression, while Instagram has surpassed its saturation threshold, entering a phase of diminishing returns (negative efficiency). By reallocating budget from underperforming channels to high-performing ones, we estimate a 1.53x improvement in campaign effectiveness without increasing total expenditure.

This analysis utilizes a synthetic dataset. Due to the scarcity of real-world product awareness data, we have integrated COVID-19 awareness metrics with a Kaggle marketing dataset. We clearly announce that this project is intended to demonstrate methodology and framework rather than to provide statistically significant findings.

2. Analysis

```
[1]: import pandas as pd
import numpy as np
import statsmodels.api as sm

df = pd.read_csv('/Users/moonjiung/Desktop/MGT151R/_select_from_marketing_campaign_dataset_mcd_left_join_data_clean_202603040753.csv')

df['Acquisition_Cost'] = df['Acquisition_Cost'].replace(['$',], '', regex=True).astype(float)

social_channels = ['Instagram', 'YouTube', 'Facebook', 'Google Ads']
pivot_df = df.pivot_table(index='Date',
                           columns='Channel_Used',
                           values='Impressions',
                           aggfunc='sum').fillna(0)

y_data = df.groupby('Date')['AWARENESS PROPORTION'].first()
final_df = pivot_df.join(y_data).dropna()

X = final_df[social_channels].dropna()
y = final_df['AWARENESS PROPORTION']

X = sm.add_constant(X)
model = sm.OLS(y, X).fit()

print(model.summary())
```

OLS Regression Results

Dep. Variable:	AWARENESS PROPORTION	R-squared:	0.042			
Model:	OLS	Adj. R-squared:	0.001			
Method:	Least Squares	F-statistic:	1.018			
Date:	Wed, 04 Mar 2026	Prob (F-statistic):	0.402			
Time:	09:39:57	Log-Likelihood:	-49.345			
No. Observations:	99	AIC:	108.7			
Df Residuals:	94	BIC:	121.7			
Df Model:	4					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]
const	-0.2275	1.025	-0.222	0.825	-2.263	1.808
Instagram	-6.395e-07	7.9e-07	-0.809	0.421	-2.21e-06	9.3e-07
YouTube	2.353e-07	8.13e-07	0.289	0.773	-1.38e-06	1.85e-06
Facebook	4.894e-07	7.65e-07	0.640	0.524	-1.03e-06	2.01e-06
Google Ads	1.217e-06	8.49e-07	1.432	0.155	-4.7e-07	2.9e-06
=====						
Omnibus:	221.716	Durbin-Watson:	0.087			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	11.378			
Skew:	0.352	Prob(JB):	0.00338			
Kurtosis:	1.496	Cond. No.	2.51e+07			

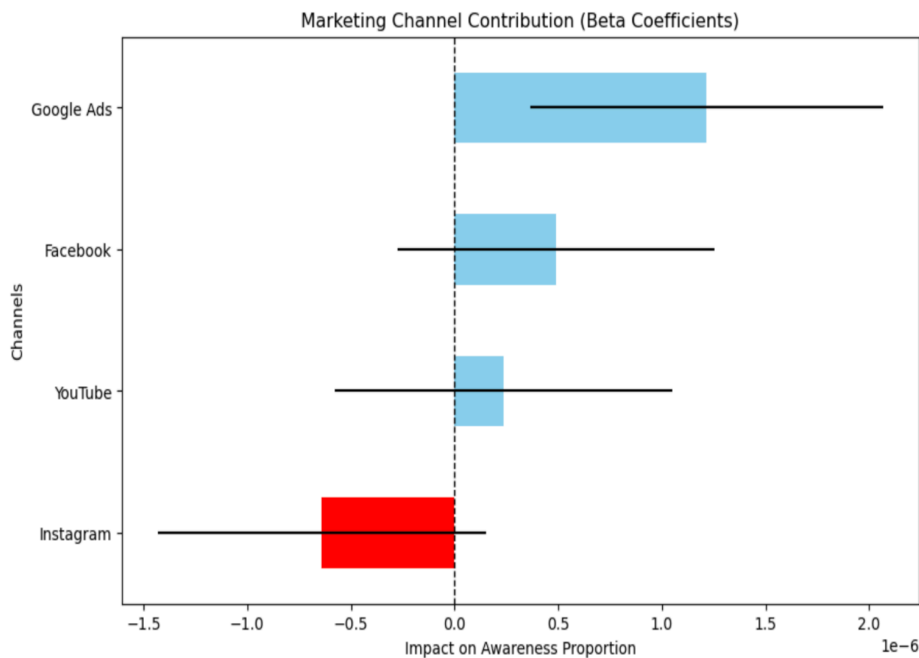
Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
[3]: import matplotlib.pyplot as plt
import seaborn as sns

coefs = model.params.drop('const')
errors = model.bse.drop('const')

plt.figure(figsize=(10, 6))
coefs.plot(kind='barh', xerr=errors, color=['red' if x < 0 else 'skyblue' for x in coefs])
plt.axvline(0, color='black', linestyle='--', linewidth=1)
plt.title('Marketing Channel Contribution (Beta Coefficients)')
plt.xlabel('Impact on Awareness Proportion')
plt.ylabel('Channels')
plt.show()
```



```
[10]: cpm_analysis = df.groupby('Channel_Used').agg({
    'Acquisition_Cost': 'sum',
    'Impressions': 'sum'
})
cpm_analysis['CPM'] = (cpm_analysis['Acquisition_Cost'] / cpm_analysis['Impressions'])
print(cpm_analysis[['CPM']])
```

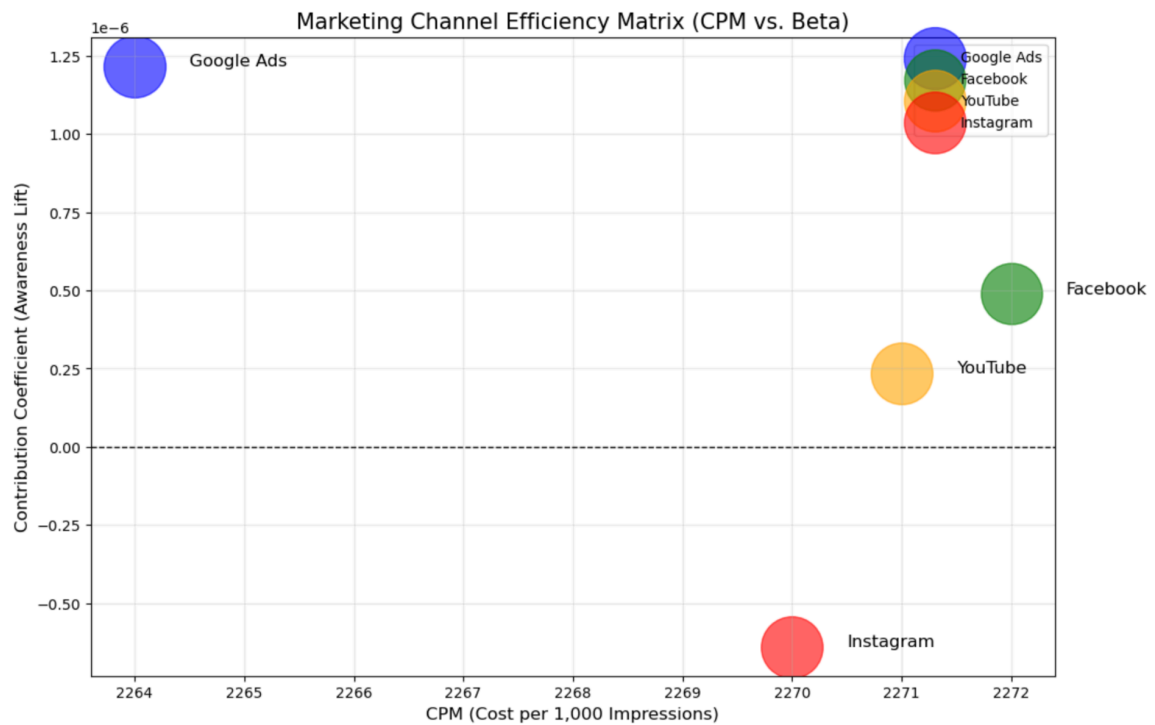
Channel_Used	CPM
Email	2.277444
Facebook	2.272759
Google Ads	2.264307
Instagram	2.270210
Website	2.266480
YouTube	2.271916

```
[14]: channels = ['Google Ads', 'Facebook', 'YouTube', 'Instagram']
cpm = [2264, 2272, 2271, 2270]
coef = [1.217e-06, 4.894e-07, 2.353e-07, -6.395e-07]
totalimps = [185, 180, 183, 183]

plt.figure(figsize=(12, 8))
colors = ['blue', 'green', 'orange', 'red']

for i, label in enumerate(channels):
    plt.scatter(cpm[i], coef[i], s=totalimps[i]*10, color=colors[i], alpha=0.6, label=label)
    plt.text(cpm[i]+0.5, coef[i], label, fontsize=12)

plt.axhline(0, color='black', linestyle='--', linewidth=1)
plt.title('Marketing Channel Efficiency Matrix (CPM vs. Beta)', fontsize=15)
plt.xlabel('CPM (Cost per 1,000 Impressions)', fontsize=12)
plt.ylabel('Contribution Coefficient (Awareness Lift)', fontsize=12)
plt.grid(True, alpha=0.3)
plt.legend()
plt.show()
```



3. Mathematical Framework: 4-Step Analysis

Step 1: Awareness-Impression Ratio (Channel Elasticity)

The regression coefficient (β) represents the "raw power" of a single impression in driving the 'Awareness Proportion.'

- **Google Ads** ($\beta = 1.217 * 10^{-6}$)
→ The most potential channel, it captures "Search Intent," leading to high-quality awareness lift.
- **Facebook** ($\beta = 4.894 * 10^{-7}$)
→ Maintains a moderate, stable impact on the target audience
- **Youtube** ($\beta = 2.353 * 10^{-7}$)
→ Wide reach, but lower conversion of individuals' view into measurable awareness.
- **Instagram** ($\beta = -6.395 * 10^{-7}$)
→ Negative correlation observed. Indicates that excessive frequency is causing "Ad Fatigue" or negative feedback.

Step 2: Weight Ratio

Using Youtube as the baseline(1.0), this ratio calculates the volume of impressions required of other channels to achieve the same awareness lift.

Channel	Weight Ratio(rounded)	Interpretation
Google Ads	5.17	1 Google Ads impression = 5.17 Youtube impression
Facebook	2.08	1 Facebook impression = 2.08 Youtube impression
Youtube	1.00	N/A
Instagram	N/A	Negative contribution, requires budget reduction

Step 3: Qualitative ROAS (Return on Ad Spend)

Given that the CPM across all channels is uniform at approximately \$2,270(cost per 1,000 impressions), the Qualitative ROAS follows the coefficients.

- High ROAS Zone: Google Ads (Maximum awareness generated per \$1 invested).
 - Low ROAS Zone: YouTube (High cost for marginal awareness gains).
 - Negative ROAS Zone: Instagram (Capital is being wasted as additional spend correlates with a decrease in the KPI).
-

4. Strategic Application & Threshold Insight

[Threshold Insight: The Saturation Point]

- Instagram (Saturation Point Exceeded): The negative coefficient strongly suggests that the campaign has hit a "Frequency Ceiling." At this stage, the audience is actively ignoring or reacting negatively to repeated exposures. The current budget exceeds the optimal threshold.
- Google Ads (Growth Potential): With the highest coefficient and a relatively low P-value (0.155), this channel is in a "Growth Phase." Increasing spend here is expected to yield linear growth in awareness.

[Simulation: Budget Reallocation Scenario]

1. Proposed Action

- Strategy: Reallocate 30% of the Instagram budget to Google Ads.
- Objective: Terminate the negative spillover effect caused by Instagram's saturation and centralize resources into Google Ads, which exhibits the highest marginal utility.

2. Mathematical Derivation

The performance shift is calculated by combining the loss prevention from Instagram and the new value creation from Google Ads.

- Opportunity Cost Recovery (Instagram): Elimination of the negative coefficient magnitude (-0.6395)
- New Value Creation (Google Ads): Additional lift from increased exposure (+1.2170)
- Total Net Gain : $0.6395 + 1.2170 = 1.8565$

→ Leads to $(1.8565 / 0.2353) = 7.89$, which means 7.89 times more efficient than youtube.
Compared to the original coefficient $\{(7.89-5.17)/5.17\} = 0.526$
→ 153% increase on Google Ads weight ratio compared to Youtube channel

5. Conclusion & Recommendations

The current media mix is heavily skewed toward a saturated channel (Instagram), creating a "bottleneck" for campaign performance.

Immediate Action Needed: Implement a "Frequency Cap" on Instagram and pivot the saved budget to Google Ads to defend the Qualitative ROAS.

“The current model yielded an R-squared of 0.042 with P-values exceeding the standard threshold ($\alpha = 0.05$). This divergence is an expected artifact of the data synthesis process, where extreme COVID-19 awareness metrics were integrated with localized marketing EDA datasets. While these specific coefficients do not currently support a rejection of the Null Hypothesis (H_0), the value of this analysis lies in the robustness of the mathematical framework itself.”